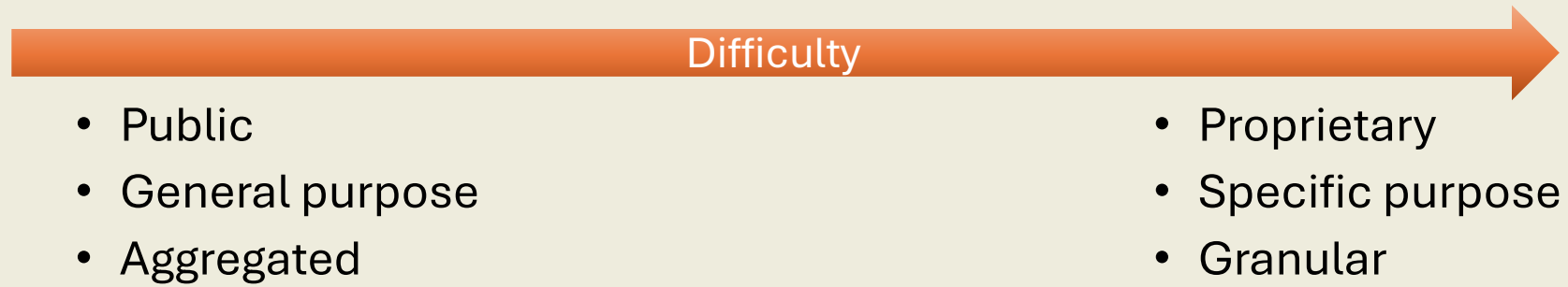


Generating and Collecting Sports Data

Saryu Onishi

HES 3000-002, Sport Performance Analytics

Data Access



WHO
COLLECTS SPORTS DATA?

League/Competition Providers

- Leagues may provide standardize data across all teams/matches
 - Examples: NBA stats, FIFA match data
- Pros:
 - Standardized
 - Easy to access
- Cons:
 - One-size-fits-all
 - Slow to innovate



Teams (Internal data)

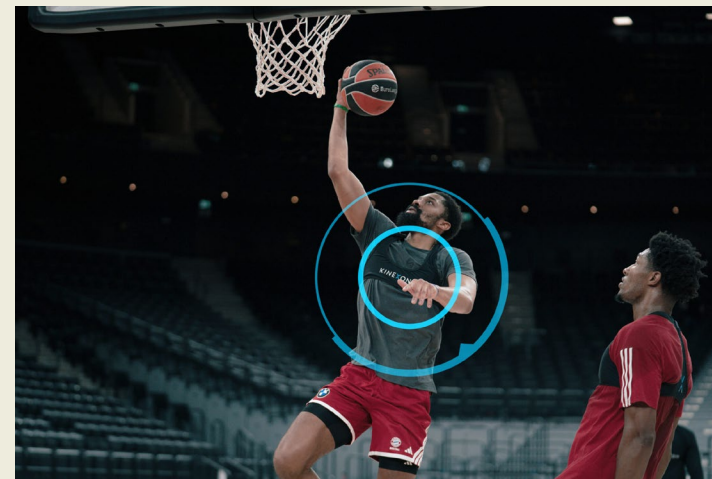
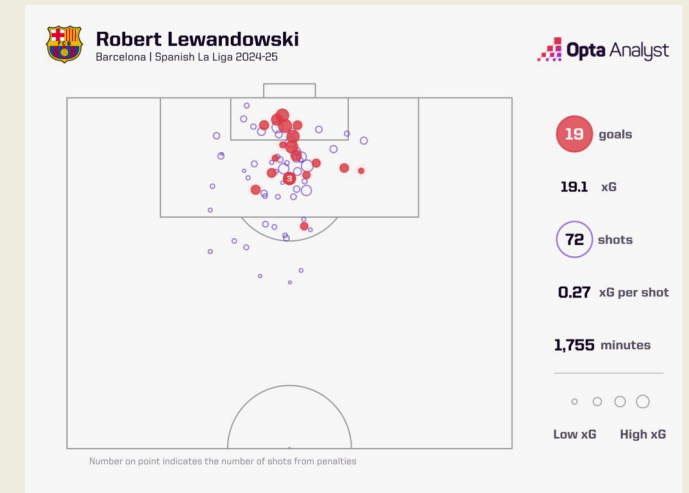
- Teams often try to collect and analyze data to gain a competitive advantage
 - Examples: Proprietary load monitoring, custom tactical metrics
- Pros:
 - Tailored to specific analytical needs
 - May create competitive advantage
- Cons:
 - No external benchmarks
 - Inaccessible to outsiders



OKLAHOMA
SPORTS SCIENCE

Third-Party Vendors

- Some businesses sell data collection technology and data analysis software
 - Opta, Stats Perform, Catapult, Kinexon, etc.
- Pros:
 - Expertise and scale
 - Access (for a price)
- Cons:
 - Black box algorithms
 - Vendor lock-in
 - Minimal competitive advantage



Academic/Research Institutions

- Research institutions prioritize methodological rigor.
- Pros:
 - Reliable methods
 - scientific collaboration
- Cons:
 - Limited samples
 - specific contexts
 - slow dissemination

HOW **IS SPORTS DATA COLLECTED?**

Observational

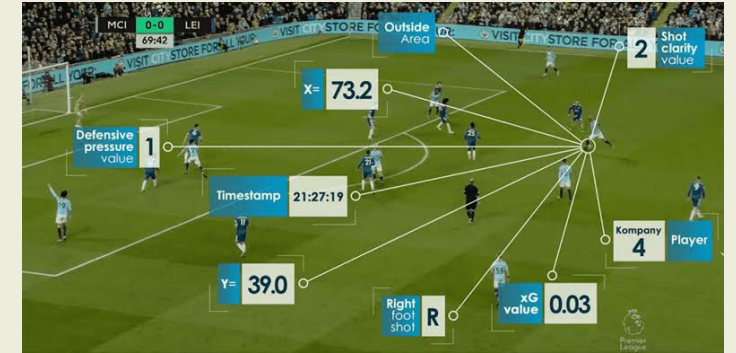
- Scout with notebook, coach's tactical board, video coding
- Strengths: Contextual, can capture nuanced behaviors
- Limitations: Schema determines possibilities, subjective, time-intensive



Commercial technologies

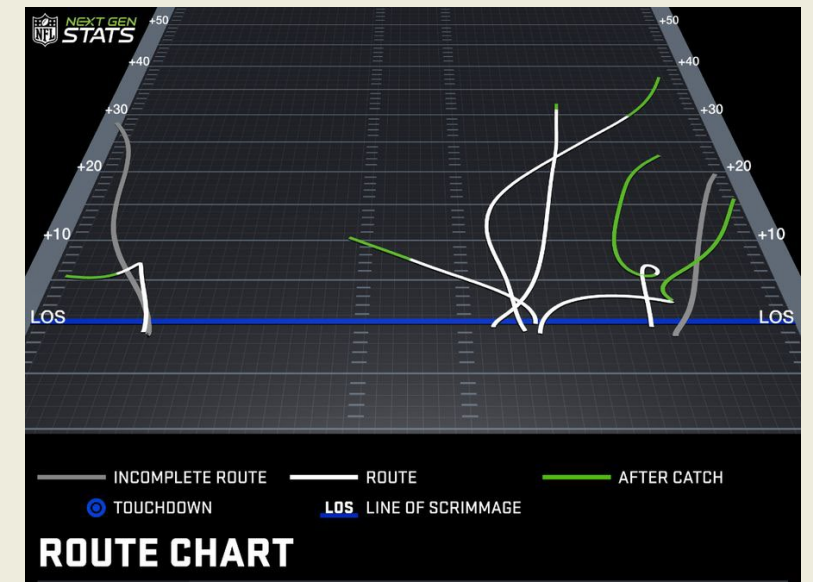
- **Optical Tracking:**

- Camera-based spatial coordinates (x, y, z).
- Non-invasive, but venue-dependent, expensive, and generates massive data volumes.
- Eg: Second Spectrum, Hawk-Eye



- **Wearables & IMUs:**

- Continuous physiological/mechanical load metrics.
- Highly specific but requires athlete buy-in
- Eg: GPS, Accelerometers, HR Monitors



Laboratory Equipment

- Force plates, motion capture, metabolic carts, isokinetic testing
- Strengths: Gold-standard precision, controlled conditions
- Limitations: Artificial environment, doesn't capture competition reality



Human vs Instrument

Human-recorded data

- Examples:
 - Scouting grades
 - Notational analysis
 - Coach observations
 - Injury or participation records
 - Event tagging

Instrument-generated data

- Examples:
 - GPS or local positioning systems
 - Optical tracking
 - Force plates
 - Heart-rate monitors
 - Timing systems

Primary vs Secondary

Primary data

- You collect data for a specific question
- Pros: Full control over what and how data are collected
- Cons: Time and resource intensive, access can be a challenge

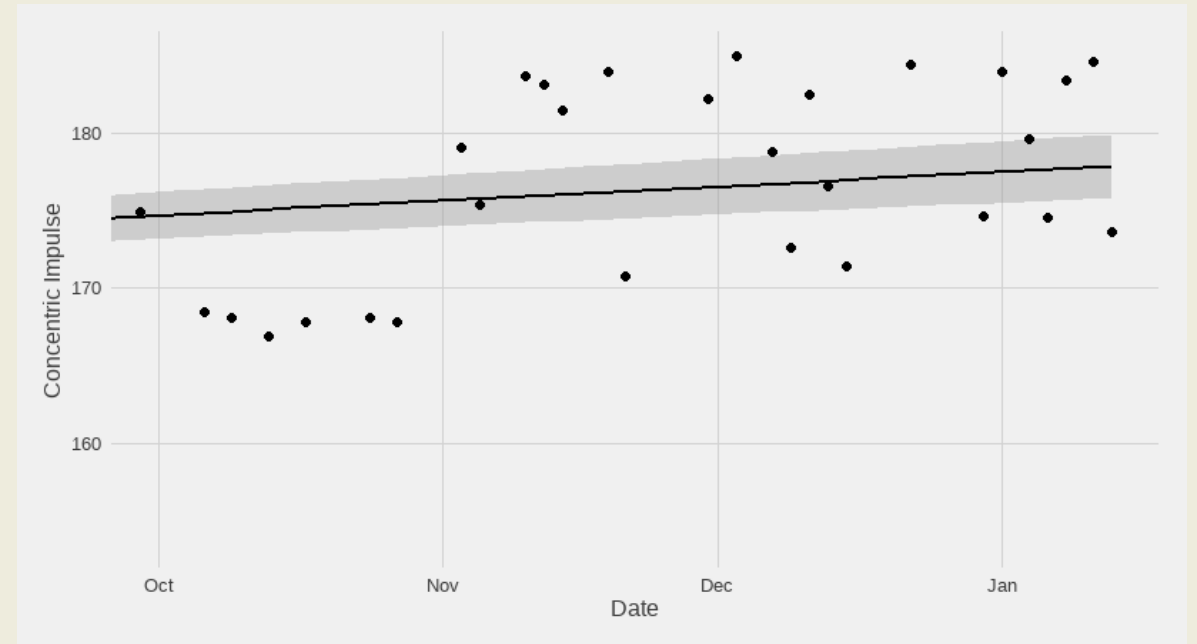
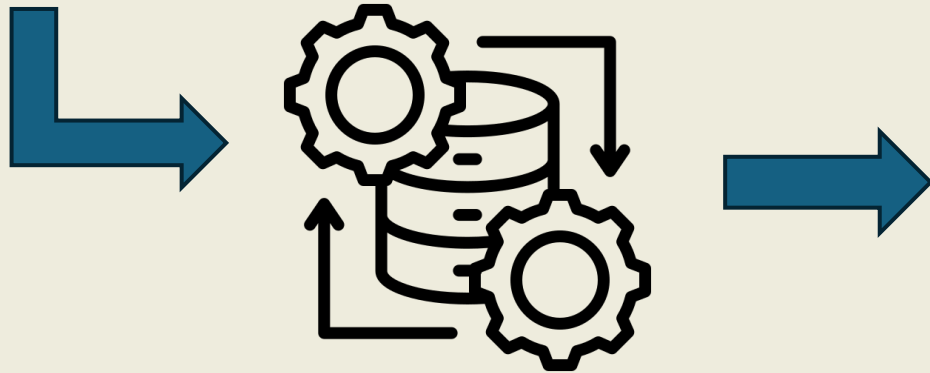
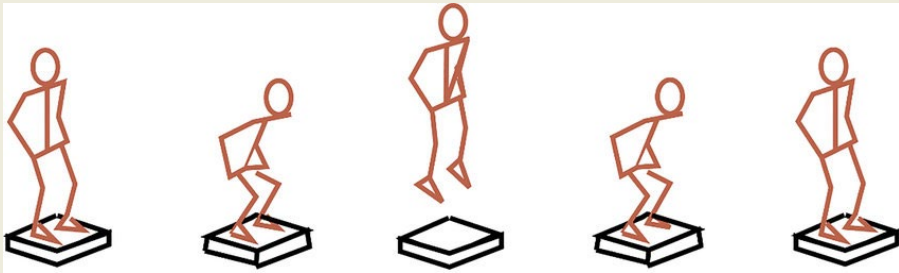
Secondary data

- Data collected by someone else for their specific question or general use case
- Pros: does not require time and resources, or direct access to sport or athlete
- Cons: may not fit your analytical question

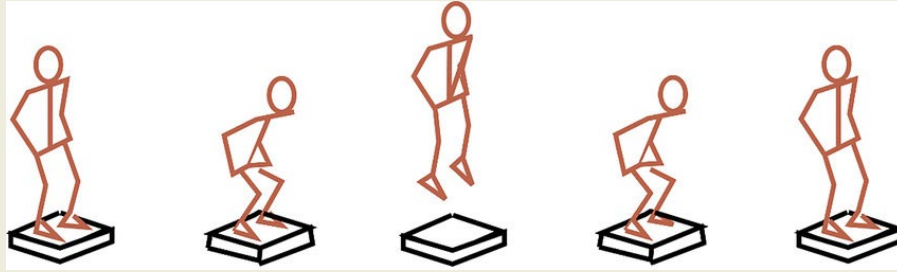
WHAT
DOES THIS DATA REPRESENT?

Event → Dataset Pipeline

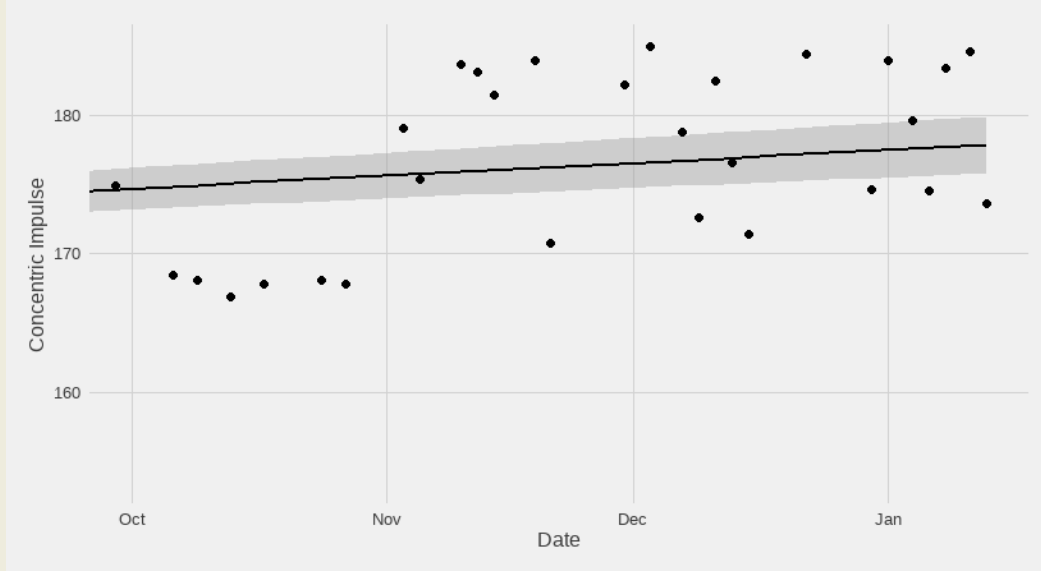
- event → data collection → data processing → analytics



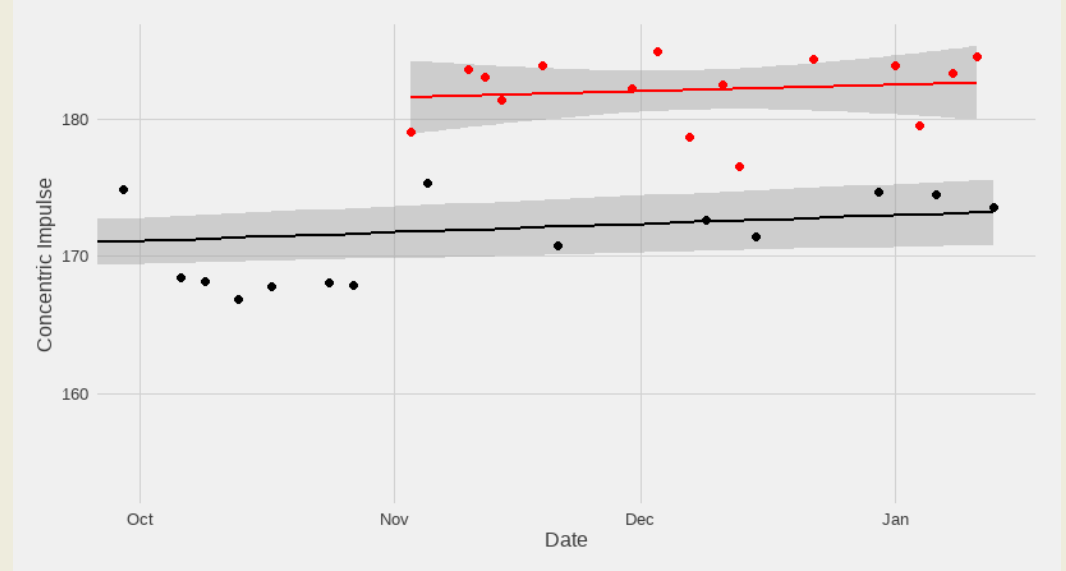
Context from the “what”



data

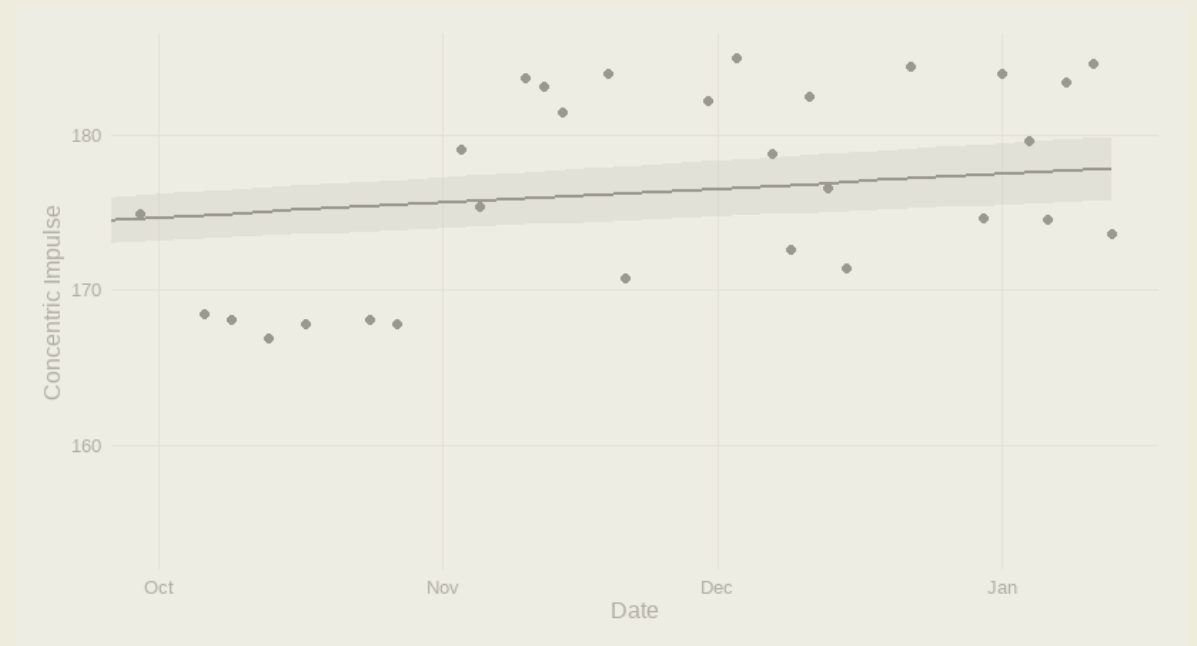
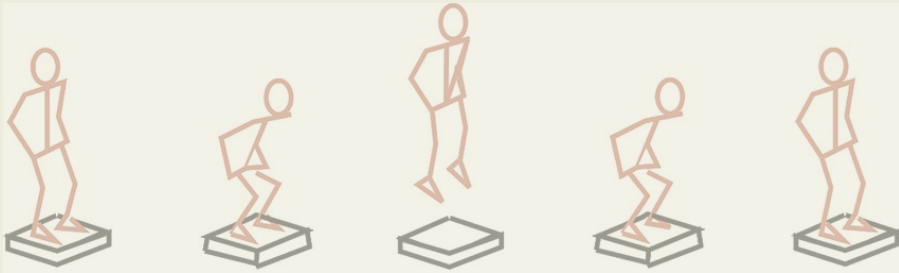


CMJ before practice and games



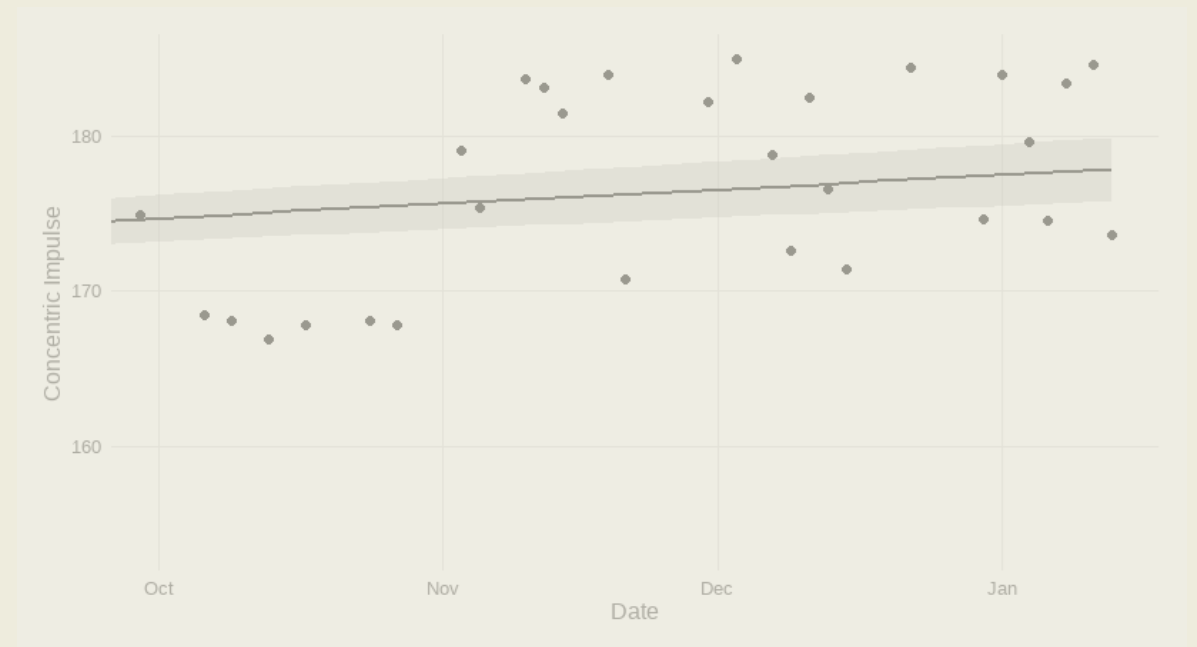
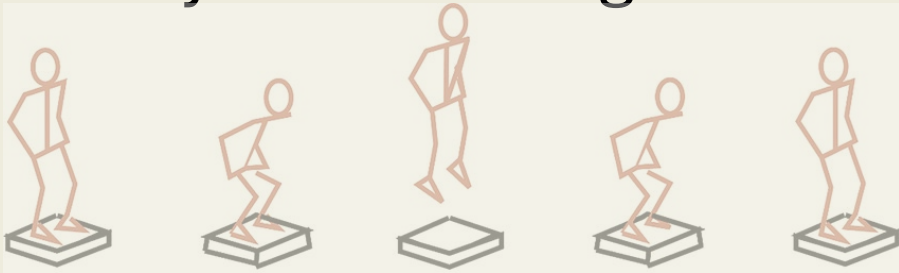
Direct / Raw Measurements

- Unprocessed or minimally processed signals straight from the sensor, human observer, or box score.
- Objective, transparent, but high-volume and often noisy.



Derived / Calculated Metrics

- Variables created by applying formulas, algorithms, aggregations, or statistical models to raw data.
- High interpretability but carries assumptions and choices made by whoever engineered the metric.



Dealing with incomplete data

- Adjust scope of question
- Be creative and find a proxy measure
 - Eg: workload data is often inaccessible. However, schedule density, back-to-back flags, or minutes played can serve as proxies.

Summary & Looking Ahead

- Know who collected the data, how the data was collected and what the data represents
- Understand how these may affect your data collection and analysis
- Next: data characteristics